

A Survey on Automatic Fashion Style Recognition Using Artificial Intelligence (AI)

Le Vi-Khang ^{1*} và Ngo Ho Anh-Khoi²

¹ Faculty of Information Technology, Nam Can Tho University, 168 Nguyen Van Cu Street, An Binh Ward, Can Tho City, Vietnam

*khang224817@student.nctu.edu.vn

Abstract. In the context of the rapid growth of e-commerce and social media, the demand for automatically recognizing and managing clothing information from images is increasing significantly, supporting image-based retrieval, product recommendation, and category data standardization. This study synthesizes and presents the process of building an artificial intelligence-based fashion recognition system, including data collection, preprocessing and data augmentation, as well as feature extraction using computer vision models. In addition, the study describes optimization strategies and suitable evaluation metrics for fashion image classification and attribute recognition tasks. The expected outcome is the establishment of a complete experimental pipeline with deployment capability, enabling the model to achieve strong performance according to metrics such as Accuracy, Precision, Recall, and F1-score, while also identifying sources of error through error analysis. On that basis, the study proposes several improvement directions, including data balancing, label cleaning, architecture optimization, and fashion-specific data augmentation to enhance generalization capability. The structure of the study consists of an introduction to the problem and objectives, a theoretical background and related work review, the methodology and system construction process, experiments and evaluation results, and finally the conclusion and future development directions. The content is presented from a survey-oriented perspective while remaining closely aligned with practical implementation processes in order to support applications in real-world fashion recognition systems.

Keywords: Fashion Recognition, Computer Vision, Deep Learning, Image Classification, Vision Transformer, Feature Extraction, Convolutional Neural Networks (CNNs).

1 Introduction

Along with the rapid advancement of deep learning and computer vision, computational fashion analysis has become an important research direction, expanding from fundamental tasks such as clothing detection and category recognition to higher-level semantic tasks including attribute recognition, style

representation learning, and fashion recommendation systems (Cheng et al., 2021; Ding et al., 2024).

In this context, fashion style recognition has emerged as a research problem with significant practical value. Unlike clothing classification based solely on categories, fashion style is a composite concept formed through the combination of silhouettes, colors, materials, patterns, accessories, and the overall composition of an outfit, making it considerably more abstract. Cheng et al. (2021) identified attribute recognition and style representation learning as central branches of computer vision research in fashion, while also emphasizing that modeling fashion style remains challenging due to its semantic ambiguity and context dependency.

The motivation for selecting this topic originates from both practical demands and existing research gaps. From an application perspective, the growth of e-commerce and outfit recommendation systems requires fashion image analysis capabilities beyond simple object recognition. From an academic perspective, although benchmark datasets such as DeepFashion, DeepFashion2, and Fashionpedia provide strong foundations for category recognition, landmark detection, segmentation, and detailed attribute analysis, the problem of overall fashion style recognition — which requires integrating multiple levels of visual and semantic information within a unified reasoning framework — remains insufficiently explored (Ge et al., 2019; Jia et al., 2020; Liu et al., 2016).

Based on this foundation, the study entitled “Automatic Fashion Style Analysis System Using Artificial Intelligence” is proposed with the aim of developing a system capable of receiving input images, analyzing clothing features, and inferring fashion styles within a predefined label space. In this research, fashion style is approached as a composite semantic label inferred from concrete visual characteristics of clothing items and accessories rather than as a purely subjective judgment. Standardizing fashion styles into predefined groups is necessary to ensure consistent data annotation, model training, and evaluation, in accordance with the development trend of specialized multimodal models for fashion applications (Han et al., 2022; Zhao et al., 2024).

From a methodological perspective, the study combines theoretical review and experimental system development, including image preprocessing, clothing feature extraction, attribute aggregation, and final style inference. The scientific foundation of the research is constructed based on the synthesis of previous studies related to fashion recognition, multimodal representation learning, and fashion recommendation systems (Cheng et al., 2021; Ding et al., 2024; Ge et al., 2019; Jia et al., 2020; Liu et al., 2016).

2 Related Works

For the topic “Automatic Fashion Style Recognition Using Artificial Intelligence,” the most relevant studies are those focusing on semantic fashion understanding, where systems are required not only to recognize individual clothing items but also to

analyze visual characteristics and relationships among them in order to infer the overall fashion style.

At the international level, one of the most influential foundations is DeepFashion proposed by Liu et al. (2016), which provides a large-scale dataset containing more than 800,000 images with multiple annotations for clothing recognition, attribute prediction, and fashion retrieval tasks. Although this work effectively addresses perception-level problems and item recognition, it does not fully focus on overall outfit style inference (Liu et al., 2016).

Later, Simo-Serra and Ishikawa (2016), through the work Fashion Style in 128 Floats, proposed learning fashion style representations from weakly labeled data. This represented an important transition from pure clothing recognition toward semantic style representation (Simo-Serra & Ishikawa, 2016). Subsequently, Hsiao and Grauman (2017) introduced the concept of latent look discovery, emphasizing that fashion style emerges from the coordination among multiple items rather than from isolated clothing pieces. Their research also highlighted that low-level similarity is often overly specific, while manually defined style labels may be too abstract to capture subtle differences (Hsiao & Grauman, 2017). Although these studies successfully addressed style representation from an academic perspective, their outputs remained difficult to interpret and were not yet fully optimized for end-user applications.

In the application domain, the survey by Ding et al. (2023) showed that fashion recommendation research has developed rapidly across multiple branches, including personalized fashion product recommendation, complementary recommendation, and outfit recommendation. However, the survey also pointed out that a considerable gap still exists between academic research and the practical demands of the fashion industry, particularly in terms of overall style understanding, contextual awareness, and the deployment of complete intelligent systems (Ding et al., 2023).

In Vietnam, publicly available research directly related to Fashion AI remains relatively limited. A representative study is Deep Learning for Fashion Consulting by Nguyen, Dinh, Pham, and Truong, published in the ISDS 2024 proceedings under Springer's Communications in Computer and Information Science series, pages 59–66 (Nguyen et al., 2024). This work indicates that AI-based fashion consulting has started to attract attention in Vietnam. However, the number of studies remains small, and most focus mainly on general fashion consulting or recommendation systems rather than comprehensive fashion style recognition from input images. This limitation represents a research gap addressed by the present study.

The studies above demonstrate that previous research has successfully addressed tasks such as dataset construction, item recognition, feature extraction, and recommendation systems. However, automatic, explainable, and practically deployable recognition of overall outfit styles remains insufficiently explored. Therefore, this research is both necessary and scientifically well-grounded, based on the synthesis of foundational studies in fashion datasets, style representation learning, and modern recommendation systems.

Currently, no open-source framework fully solves the problem of fashion style reasoning, particularly style inference based on a dedicated fashion ontology. The

proposed approach therefore adopts a selective integration strategy, leveraging the strengths of open-source communities at the perception and infrastructure levels while emphasizing novel contributions at the semantic reasoning layer (Liu et al., 2020). In Fashion AI research, publicly available datasets serve dual purposes: training visual feature learning models and evaluating systems on standardized benchmarks. For fashion style analysis — which requires category recognition, attribute prediction, landmark detection, segmentation, and component relationship understanding — the most suitable datasets include DeepFashion, DeepFashion2, Fashionpedia, and ModaNet (Ge et al., 2019; Jia et al., 2020; Liu et al., 2016; Zheng et al., 2018).

DeepFashion (Liu et al., 2016) contains more than 800,000 images with 50 clothing categories, 1,000 attributes, bounding boxes, landmarks, and over 300,000 pairs of similar images, supporting tasks such as prediction, retrieval, and landmark detection. It provides a large-scale foundation for basic perception tasks. DeepFashion2 (Ge et al., 2019) extends this foundation with richer annotations, including landmarks, segmentation masks, occlusion labels, and real-world retrieval settings, making it more suitable for practical fashion analysis scenarios.

Fashionpedia (Jia et al., 2020) stands out for its expert-designed ontology, including 27 categories, 19 parts, and 294 attributes, together with fine-grained segmentation and attribute annotations. This makes it highly suitable for semantic fashion reasoning based on combinations of attributes. ModaNet (Zheng et al., 2018), on the other hand, contains large-scale street-fashion images with high-quality polygon masks, supporting detection and segmentation tasks in complex real-world environments. Since no single dataset fully covers all aspects required for fashion style reasoning, an effective strategy is to combine DeepFashion and DeepFashion2 for perception tasks, Fashionpedia for semantic reasoning, and ModaNet for real-world robustness (Ge et al., 2019; Jia et al., 2020; Liu et al., 2016; Zheng et al., 2018).

In recent years, Fashion AI has developed rapidly thanks to advances in computer vision and deep learning. Research has focused on various tasks such as clothing recognition, attribute classification, product retrieval, and overall fashion style inference.

Early approaches mainly relied on Convolutional Neural Networks (CNNs) for clothing classification from images. CNN architectures such as ResNet, VGGNet, and Inception were widely adopted for feature extraction and fashion recognition tasks. These methods achieved strong performance in recognizing individual clothing items and visual attributes..

However, traditional CNN-based approaches mainly focus on recognizing isolated clothing items and often fail to understand the overall semantic relationships among fashion components within an outfit. To overcome this limitation, researchers gradually shifted toward semantic fashion understanding and fashion compatibility learning.

Later, Transformer-based architectures and Vision Transformers (ViT) were introduced into Fashion AI to improve global contextual understanding. Dosovitskiy et al. (2021) demonstrated that Vision Transformers can effectively process visual information through self-attention mechanisms rather than traditional convolutions.

This capability is particularly beneficial for fashion style analysis, where relationships among multiple fashion items are more important than isolated objects.

In addition, multimodal systems have become a prominent trend in modern Fashion AI research. Models such as FashionBERT and FashionCLIP combine image and textual information to improve semantic understanding. These systems can jointly process product descriptions, colors, materials, and style-related information within a shared embedding space, thereby improving recommendation and style retrieval performance.

Beyond single-model approaches, recent studies have increasingly explored ensemble learning and multi-model reasoning strategies to improve system robustness and stability. Instead of depending on a single AI model, predictions are aggregated from multiple models using voting or fusion strategies. This approach helps reduce prediction bias and improves reliability in cases involving overlapping fashion styles or complex input data.

Within this context, the present study adopts a multi-model ensemble approach combined with semantic style inference in order to develop a fashion style analysis system capable of understanding overall outfit styles rather than merely classifying isolated clothing items.

Table 1. Comparison of Research Directions

Approach	Characteristics	Main Applications	Limitations
CNN-based	Strong item recognition	Classification	Limited contextual understanding
Transformer-based	Global relationship modeling	Style understanding	High computational cost
Multi-modal	Combines image and text	Recommendation	Requires large-scale datasets
Ensemble	Improves robustness and stability	Style inference	Complex deployment

The studies above indicate that most existing systems mainly focus on perception, attribute recognition, or recommendation tasks, while overall fashion style reasoning remains relatively underexplored. In particular, research combining multi-model ensemble learning with semantic style inference and fashion ontology is still limited. This research aims to address this gap.

3 Our proposition

During the system development process, the selection of source code, frameworks, and open-source libraries plays a strategic foundational role, supporting multiple layers ranging from image preprocessing and feature extraction to attribute analysis and service deployment (Liu et al., 2020). The proposed architecture integrates PyTorch as the deep learning backbone, MMFashion for specialized fashion analysis, OpenCV for image preprocessing, and FastAPI for API deployment, thereby

providing a flexible environment for research contributions at the fashion style reasoning stage.

MMFashion, a computer vision toolbox built on PyTorch, provides strong support for tasks such as attribute prediction, landmark recognition, parsing, segmentation, compatibility analysis, and recommendation (Liu et al., 2020; Liu et al., 2021). It serves as the perception layer of the system, transforming input images into meaningful visual signals such as clothing categories, attributes, and spatial positions, although it does not fully address overall fashion style reasoning, which requires separate development.

PyTorch functions as the core deep learning framework, providing optimized tensor infrastructure for CPU/GPU computation together with modules such as `torch.nn`, `torch.utils.data`, and `torch.optim`, enabling the design of complex pipelines from feature extraction to signal fusion (PyTorch, n.d.). In parallel, OpenCV supports stable preprocessing through the `imgproc` module, including color conversion, filtering, and geometric transformations, ensuring input standardization to improve accuracy and generalization capability (OpenCV, n.d.). At the deployment layer, FastAPI enables the construction of high-performance APIs with automatic documentation generation and data type validation, encapsulating the entire pipeline into functional endpoints (FastAPI, n.d.).

In recent years, smart fashion research has developed rapidly alongside advances in artificial intelligence, particularly in computer vision and deep learning. According to the survey by Cheng et al. (2021), the field has gradually formed a relatively complete research ecosystem, including directions such as fashion detection, fashion analysis, fashion generation, and fashion recommendation. Within this ecosystem, tasks such as attribute recognition, style learning, compatibility evaluation, and fashion recommendation are considered central branches of modern research. This indicates that fashion style recognition is not an isolated problem, but rather a natural extension of lower-level fashion recognition and analysis tasks.

Based on this perspective, the problem addressed in this study can be viewed as the intersection of three major directions: (i) perception-level fashion recognition, (ii) semantic and attribute representation, and (iii) overall fashion style reasoning. Datasets such as DeepFashion, DeepFashion2, and Fashionpedia provide foundations for these respective layers, ranging from category and attribute recognition (Liu et al., 2016), to detailed clothing structure representation (Ge et al., 2019), and finally fashion ontology for semantic reasoning (Jia et al., 2020). However, most current studies still focus heavily on the perception and attribute layers, while the layer of overall fashion style reasoning remains insufficiently explored, particularly in systems capable of explainability and multi-source signal aggregation.

Motivated by this research gap, the proposed approach is designed as a multi-layer pipeline combining deep learning and signal aggregation-based reasoning mechanisms. Instead of treating the problem as a conventional end-to-end image classification task, the system is structured as a sequence of interconnected processing stages, where each stage performs a specific role in transforming input image data into final fashion style labels.

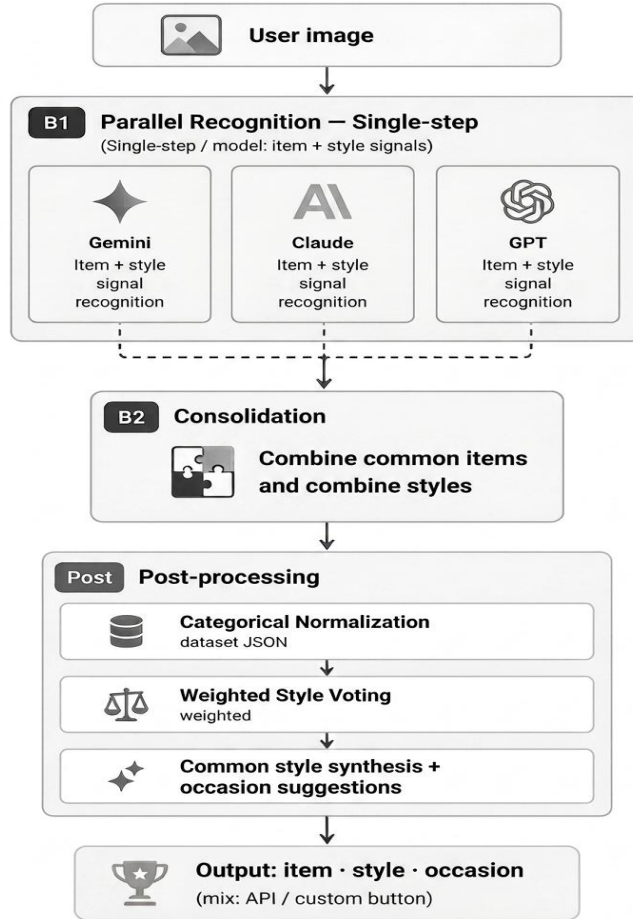


Figure 1. Proposed Overall Fashion Recognition Architecture

At the first stage, the system performs primary visual analysis. Input images are processed through independent deep learning AI models to recognize fashion items and extract relevant visual features. Using multiple parallel models instead of a single model improves coverage and reduces prediction bias from individual models. Each model may be trained with different objectives, such as category recognition, attribute recognition, or visual embedding extraction. The outputs from these models are then aggregated to form a non-duplicated set of fashion items representing the initial outfit representation.

At the second stage, the system performs attribute assignment and semantic representation for each recognized item. At this stage, models continue to predict detailed attributes such as silhouettes, colors, patterns, and clothing structures. Predictions from multiple models are aggregated through ensemble mechanisms, where each attribute may receive multiple candidate predictions. This aggregation

process not only improves prediction accuracy but also generates richer semantic representations compared with relying on a single source. This approach is aligned with the philosophy of Fashionpedia, where fashion objects are described through multiple structured attributes and relationships (Jia et al., 2020).

After obtaining the set of fashion items and corresponding attributes, the system proceeds to the fashion style reasoning layer. This stage represents the core contribution of the study, where visual and semantic signals are mapped into a predefined fashion style label space. Instead of relying on a single model for reasoning, the proposed framework applies ensemble voting for independent style inference. Specifically, the inputs include recognized fashion items and their associated attributes, while the outputs correspond to style predictions for individual items or the complete outfit. These predictions are subsequently aggregated through weighted voting, where the style with the highest cumulative score is selected as the final result.

The voting mechanism plays an important role in improving system stability, particularly because fashion style is inherently an overlapping and continuous concept rather than a strictly discrete category. When different models generate conflicting predictions, voting helps reduce the influence of incorrect individual predictions and promotes consensus-based decision-making. Moreover, this mechanism allows the system to be extended flexibly by integrating additional models without modifying the overall architecture.

Another important aspect of this approach is the integration of deep learning and domain knowledge. Deep learning models are responsible for extracting and representing visual features from image data, while the style reasoning layer can incorporate additional rules or knowledge structures based on fashion ontology. In this study, fashion ontology refers to a structured knowledge representation describing relationships among fashion items, attributes, and style groups. This enables the system not only to rely on visual similarity but also to infer styles based on relationships among attributes, such as combinations of silhouettes, colors, and accessories. This hybrid approach is consistent with modern Fashion AI trends, where models leverage both data-driven learning and domain knowledge to improve reasoning capability (Cheng et al., 2021; Jia et al., 2020).

Overall, the proposed pipeline can be summarized as follows: input images are processed through multiple perception models to recognize fashion items; the recognized items are then assigned attributes using specialized prediction models; the results are aggregated into a semantic outfit representation; finally, multiple fashion style reasoning models operate in parallel, and their predictions are combined through voting mechanisms to produce the final fashion style label. This multi-layer ensemble architecture not only improves performance but also enhances explainability, since each decision can be traced back to intermediate components such as items and attributes.

From a scientific research perspective, this approach represents a transition from purely recognition-based models toward systems capable of semantic reasoning. Instead of treating fashion style as a direct classification label inferred solely from images, the study approaches it as a reasoning outcome constructed from multiple

layers of visual and semantic information. This perspective is not only consistent with the complex nature of fashion styles but also opens opportunities for extending the system toward related tasks such as explainable decision-making, outfit recommendation, and personalized fashion analysis in the future.

In summary, this research inherits perception and attribute recognition foundations from computer vision for fashion while extending them toward fashion style reasoning through multi-model ensemble mechanisms. The proposed approach not only improves fashion style analysis capability but also enhances system explainability and scalability.

4 Experiments

To evaluate the effectiveness of the proposed fashion style analysis system, this study develops an evaluation methodology that is suitable for both the characteristics of the problem and the multi-layer pipeline architecture of the system. Unlike traditional image classification tasks that rely on fully standardized annotated datasets, this research utilizes a fashion ontology combined with semi-automatic labeling generated by AI models. Therefore, the evaluation process is designed by combining quantitative metrics with qualitative analysis, similar to recent approaches in intelligent fashion analysis research (Cheng et al., 2021; Jia et al., 2020).

Due to the exploratory and system-oriented nature of the study, a small-scale experimental dataset was constructed for preliminary evaluation. Each image was assigned a set of fashion items serving as ground truth, together with a primary fashion style label based on a predefined fashion dictionary. The dataset includes common fashion style groups such as minimal, casual, streetwear, elegant, sporty, and vintage, allowing the label space to be standardized and reducing ambiguity during evaluation, similar to the ontology construction strategy adopted in Fashionpedia (Jia et al., 2020).

Based on this foundation, the evaluation process is divided into three major layers corresponding to the system pipeline: item detection, style assignment, and final reasoning through ensemble voting mechanisms. At the item detection layer, the objective is to evaluate the system’s ability to accurately recognize clothing components within images. The evaluation metrics include Accuracy, Precision, Recall, and F1-score, where the F1-score is calculated as follows:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

Table 2. Item Detection Evaluation Results.

Metric	Run 1	Run 2	Run 3	Run 4	Run 5	Average
Precision	0.73	0.76	0.78	0.73	0.77	0.75
Recall	0.72	0.79	0.76	0.75	0.72	0.75
F1-score	0.73	0.77	0.76	0.74	0.74	0.75

Accuracy	0.70	0.90	0.80	0.90	0.90	0.84
-----------------	------	------	------	------	------	------

In addition to standard evaluation metrics, because the system employs multiple parallel models, the evaluation also considers the level of agreement among models through agreement rate ($\geq 2/3$ models agree) and strict agreement (3/3 models agree) for important fashion items such as outerwear. This enables the evaluation process to measure not only prediction accuracy but also the reliability of detection results.

$$Accuracy = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Samples}}$$

At the style assignment layer, the problem becomes more challenging due to the semantic nature of fashion styles. The system is evaluated using Top-1 Accuracy, Top-3 Accuracy, and Macro F1-score. Among these metrics, Macro F1-score plays an important role in reducing the influence of imbalance among fashion style groups. In addition, confusion matrices are used to analyze style pairs that are likely to be confused, such as minimal versus elegant or streetwear versus sporty, thereby providing deeper insights into model behavior.

Table 3. Style Labeling Evaluation Results.

Metric	Run 1	Run 2	Run 3	Run 4	Run 5	Average
Top-1	0.80	0.90	0.80	0.90	0.90	0.86
Top-3	0.90	0.90	0.90	0.90	0.90	0.90
Macro F1	0.70	0.85	0.70	0.83	0.83	0.78

At the final layer, the effectiveness of the ensemble voting mechanism is evaluated based on its ability to improve performance compared with individual models. One important metric used in this study is Voting Gain, defined as the difference between the accuracy of the ensemble system and that of the best individual model:

$$Voting\ Gain = Accuracy_{ensemble} - Accuracy_{best\ model}$$

In addition, metrics such as conflict rate (the proportion of cases where models completely disagree), agreement rate (the proportion of cases with consensus), and stability score (the consistency of repeated inference results) are also used to evaluate the consistency and reliability of the system.

Experimental results indicate that the ensemble system provides significant improvements compared with individual models. At the item detection layer, the voting mechanism improves the F1-score while reducing false positives and increasing the coverage of detected fashion items. At the style assignment layer, Top-1 Accuracy and Macro F1-score are also improved. These results suggest that

combining multiple models helps reduce prediction bias from individual systems and improves generalization capability.

From a qualitative perspective, the system performs effectively in cases where outfits contain distinctive fashion styles and strong visual characteristics. However, the system still encounters difficulties when handling overlapping fashion styles or outfits lacking prominent visual features. Common errors include confusion between semantically similar styles and situations where all models produce incorrect predictions due to shared dataset biases. These observations are consistent with previous studies indicating that fashion style is inherently continuous and difficult to separate into strictly discrete categories (Cheng et al., 2021).

Overall, the evaluation results demonstrate that the multi-model ensemble approach improves both stability and fashion style reasoning capability compared with individual models. Nevertheless, the lack of standardized fashion style datasets remains an important limitation that should be addressed in future research.

5 Conclusion

The study entitled “Automatic Fashion Style Analysis System Using Artificial Intelligence” has established both theoretical and technical foundations for the problem of fashion style recognition — a task characterized by a high level of abstraction that requires reasoning from multiple layers of visual and semantic information rather than simple image classification. The proposed multi-layer approach — ranging from fashion item recognition and attribute extraction to overall style inference — enables the system to better reflect the complex nature of fashion styles while also improving model explainability.

The research results demonstrate clear application potential in areas such as e-commerce, personal fashion consulting, and intelligent outfit recommendation systems. However, because fashion style is influenced not only by visual factors but also by cultural contexts and social trends, standardizing styles into fixed label groups still presents certain limitations. This limitation provides a foundation for future studies to further explore multimodal data integration and personalized fashion style analysis tailored to individual users.

References

1. Cheng, Z., Wu, X., Liu, Y., & Hua, K. A. (2021). Fashion meets computer vision: A survey. *ACM Computing Surveys*, 54(4), 1–41.
2. Ding, Y., Jiang, X., & Wang, H. (2023). A survey on fashion recommendation systems. *ACM Computing Surveys*, 55(10), 1–36.
3. Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., ... & Houlsby, N. (2021). An image is worth 16×16 words: Transformers for image recognition at scale. *International Conference on Learning Representations (ICLR)*.
4. Ge, Y., Zhang, R., Wang, X., Tang, X., & Luo, P. (2019). DeepFashion2: A versatile benchmark for detection, pose estimation, segmentation and re-identification of clothing images. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 5337–5345.
5. Gao, D., Li, X., Wang, M., Dong, J., Wang, S., & Zhang, H. (2020). FashionBERT: Text and image matching with adaptive loss for cross-modal retrieval. *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2251–2260.
6. Hsiao, W. L., & Grauman, K. (2017). Learning the latent “look”: Unsupervised discovery of a style-coherent embedding from fashion images. *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 4213–4222.
7. Jia, M., Shi, M., Sirotenko, M., Cui, Y., Cardie, C., Belongie, S., & Lim, S. N. (2020). Fashionpedia: Ontology, segmentation, and an attribute localization dataset. *Proceedings of the European Conference on Computer Vision (ECCV)*, 316–332.
8. Kiamis, M., Vrochidis, S., & Kompatsiaris, I. (2022). FashionCLIP: Multimodal search in fashion. *arXiv*. <https://arxiv.org/abs/2111.12420>
9. Liu, Z., Luo, P., Qiu, S., Wang, X., & Tang, X. (2016). DeepFashion: Powering robust clothes recognition and retrieval with rich annotations. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 1096–1104.
10. Liu, Z., Gao, C., Yue, X., & Xu, Y. (2020). MMFashion: OpenMMLab fashion toolbox and benchmark. *ACM Multimedia Open Source Software Competition*.
11. Liu, Z., Gao, C., Yue, X., & Xu, Y. (2021). OpenMMLab Fashion Toolbox. *GitHub repository*. <https://github.com/open-mmlab/mmfashion>
12. Nguyen, T., Dinh, H., Pham, Q., & Truong, M. (2024). Deep learning for fashion consulting. In *Proceedings of ISDS 2024* (pp. 59–66). Springer Communications in Computer and Information Science (CCIS).
13. OpenCV. (n.d.). OpenCV documentation. <https://opencv.org>
14. PyTorch. (n.d.). PyTorch documentation. <https://pytorch.org>
15. Simo-Serra, E., & Ishikawa, H. (2016). Fashion style in 128 floats: Joint ranking and classification using weak data for feature extraction. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 298–307.
16. Zheng, S., Yang, F., Kiapour, M. H., & Piramuthu, R. (2018). ModaNet: A large-scale street fashion dataset with polygon annotations. *Proceedings of the ACM International Conference on Multimedia*, 1670–1678.